Causes of Histamine Buildup in the Blood

Histamine buildup occurs when histamine release exceeds the body's ability to break it down or clear it. Histamine is a natural chemical involved in allergic reactions, immune defence, stomach acid secretion and nervous system signalling.

1. Major causes of increased histamine activity

A. Histamine intolerance and reduced breakdown

The enzyme diamine oxidase (DAO) mainly breaks down histamine from food in the intestine, while histamine-N-methyltransferase (HNMT) helps metabolise histamine inside cells. Reduced breakdown of dietary histamine may contribute to symptoms in some people. However, histamine intolerance has no universally accepted diagnostic standard.

B. Foods that contain histamine

Histamine levels can be higher in foods that are aged, fermented, cured or stored improperly.

Food category	Examples
Fermented foods	Yoghurt, sauerkraut, fermented vegetables
Aged foods	Mature cheeses, cured meats
Fish	Smoked or canned fish; poorly refrigerated fish
Alcoholic drinks	Red wine, beer
Selected vegetables	Tomatoes, spinach, eggplant
Other foods	Vinegar-containing or some processed foods

Some foods are described as "histamine liberators," but evidence that they directly trigger histamine release is limited.

C. Allergic reactions

Allergies can cause mast cells and basophils to release histamine. Possible triggers include pollen, dust mites, animal dander, certain foods, medications and insect stings. Symptoms can include itching, hives, sneezing, swelling and breathing difficulties.

D. Mast cell disorders

Mast cell activation syndrome (MCAS) and mastocytosis can involve excessive release of histamine and other inflammatory chemicals. Symptoms may include flushing, repeated hives, diarrhoea, dizziness and episodes of low blood pressure.

E. Digestive and intestinal conditions

Coeliac disease, inflammatory bowel disease, intestinal infections and other causes of gut inflammation may affect handling of dietary histamine. Gut bacteria may also influence histamine production, although the clinical implications are still being studied.

F. Certain medications

Some opioid pain medicines, antibiotics or anti-inflammatory medications can affect histamine pathways or cause mast-cell-mediated reactions. The effect depends on the individual medicine. Do not stop a prescribed treatment without medical advice.

2. Symptoms that may accompany excessive histamine activity

- **Skin:** Itching, hives, redness, flushing or swelling.
- **Digestive system:** Bloating, abdominal cramps, nausea or diarrhoea.
- **Heart and circulation:** Palpitations, fast heartbeat, dizziness or low blood pressure.
- **Head and respiratory system:** Headaches, nasal congestion, sneezing or wheezing.

These symptoms have many possible explanations. They do not, by themselves, establish that histamine is elevated.

3. Tests and medical assessment

- **Allergy testing:** Helps assess suspected specific allergens.
- **Serum tryptase:** Can help evaluate suspected mast cell activation, sometimes using samples taken during an episode and at baseline.
- **Urinary histamine metabolites:** May be considered by specialists assessing mast cell disorders.
- **Assessment for intestinal conditions:** Considered when significant gastrointestinal symptoms are present.
- **Blood histamine:** Results are difficult to interpret because levels change quickly and are affected by sample handling.

A single high blood histamine result does not diagnose MCAS or histamine intolerance. DAO testing alone is also not a reliable stand-alone diagnostic test.

4. Ways to manage histamine-related symptoms

- Identify patterns with a food-and-symptom diary and avoid known allergens.
- Treat confirmed underlying allergies or digestive conditions.
- Consider antihistamines only when appropriate for your symptoms and medical history.
- A clinician or dietitian may suggest a supervised, short-term low-histamine food trial if symptoms repeatedly follow meals; avoid unnecessarily restrictive long-term diets.

Emergency warning: Sudden breathing difficulty, swelling of the tongue or throat, or collapse/fainting may indicate anaphylaxis. Seek emergency medical care immediately.

5. Questions to discuss with a healthcare professional

- What symptoms occur: itching, skin rash, headache, palpitations, abdominal discomfort or others?
- Has a laboratory test shown high histamine, or is histamine suspected because of symptoms?
- Do symptoms follow particular meals, medications or exposures, or occur throughout the day?

This document is educational and does not replace medical diagnosis or treatment.